

Computational Geometric Approach for BIM Semantic Enrichment to Support Automated Underground Garage Compliance Checking

Han Gao¹; Botao Zhong²; Hanbin Luo³; and Weiya Chen, Ph.D.⁴

Abstract: The need for manual semantic annotation to domain-specific building information modeling (BIM) models limits the application of BIM technology in various architecture, engineering, and construction fields. BIM semantic enrichment refers to a procedure that can identify missing information and improve interoperability across different applications. The need for semantic enrichment of BIM models is often emphasized in the literature; however, few studies have shown the potential of semantic enrichment approaches in dealing with problems in real-world cases. To provide an empirical demonstration in the practice, an example of BIM semantic enrichment for supporting underground garage compliance checking is selected as a case study. Geometric reasoning is one of the effective approaches to enhance semantic information in BIM models to support real-world applications. In this study, a computational geometric approach is developed to demonstrate how geometric information can be extracted and used to infer missing semantics in underground garage BIM models. A prototype system is developed, and real underground garage BIM models are used to validate the effectiveness of the proposed approach. The contribution of this study is that it provides the detailed implementation of performing geometric reasoning to enhance BIM semantics in real-world cases. Additionally, the findings of this case study show that the proposed computational geometric approach has the potential to enhance the semantics of BIM models to support decision making in real-world engineering problems, and the developed algorithms can be used to automate the compliance checking process of the underground garage. DOI: [10.1061/\(ASCE\)CO.1943-7862.0002230](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002230). © 2021 American Society of Civil Engineers.

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Introduction

In recent decades, BIM technology has been applied in many fields in the architecture, engineering, and construction (AEC) industry, such as code checking (Martins and Monteiro 2013; Choi et al. 2014; Ji and Leite 2018), energy analysis (Ying and Lee 2019), and safety risk identification (Hossain et al. 2018). Most of these

applications are supported by the Industry Foundation Class (IFC) (Bloch and Sacks 2018). IFC is an open data schema that integrates various relevant information for construction projects (Malsane et al. 2015). The IFC model has very rich entities and attributes, but it cannot cover every specific domain. The main challenge for existing BIM automation applications is the semantic mismatch between the IFC model and the specific AEC domain. In many design review applications, additional attributes and relationships need to be pre-defined to respond to rule queries before executing the code checking program (Eastman et al. 2009). BIM semantic enrichment has provided an alternative way to fill the gap between the BIM models and the required information by the user. BIM semantic enrichment is a procedure wherein domain expert knowledge is used to infer new objects, attributes, and spatial relationships from a BIM model. The missing information in the IFC schema can be identified to support applications in the AEC field (Sacks et al. 2017).

Semantic enrichment of BIM models is considered an emerging field of research, and Bloch and Sacks have classified BIM semantic enrichment tasks as classification of building objects, unique identification, completion of missing objects, and reconstruction of occluded objects. (Bloch and Sacks 2018). Various types of approaches (such as machine learning methods and geometric reasoning methods) have been developed to enhance the semantics in BIM models. For example, machine learning methods such as support vector machines (Koo et al. 2019; Bloch and Sacks 2018) and 3D deep neural networks (Koo et al. 2021) are used to extract information from BIM models to classify or identify BIM elements. Except for the machine learning methods, geometric reasoning methods have also been developed to deal with elements classification problems such as the classification of bridge objects (Ma et al. 2018).

¹Ph.D. Candidate, Dept. of Construction Management, School of Civil Engineering and Mechanics, Huazhong Univ. of Science and Technology, Wuhan 430074, Hubei, China; Ph.D. Candidate, Hubei Engineering Research Center for Virtual, Safe and Automated Construction, Wuhan 430074, Hubei, China. Email: ghandhunter@163.com

²Professor, Dept. of Construction Management, School of Civil Engineering and Mechanics, Huazhong Univ. of Science and Technology, Wuhan 430074, Hubei, China; Ph.D. Candidate, Hubei Engineering Research Center for Virtual, Safe and Automated Construction, Wuhan 430074, Hubei, China (corresponding author). Email: dadizhong@hust.edu.cn

³Professor, Dept. of Construction Management, School of Civil Engineering and Mechanics, Huazhong Univ. of Science and Technology, Wuhan 430074, Hubei, China; Ph.D. Candidate, Hubei Engineering Research Center for Virtual, Safe and Automated Construction, Wuhan 430074, Hubei, China. Email: luohbcm@hust.edu.cn

⁴Dept. of Construction Management, School of Civil Engineering and Mechanics, Huazhong Univ. of Science and Technology, Wuhan 430074, Hubei, China; Ph.D. Candidate, Hubei Engineering Research Center for Virtual, Safe and Automated Construction, Wuhan 430074, Hubei, China. Email: weiya_chen@hust.edu.cn

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and structural elements (Wu and Zhang 2019). Most literature on semantic enrichment is dedicated to deal with only one aspect of the problems such as classification and identification. However, few empirical demonstrations exist that show the potential of approaches of semantic enrichment in real-world cases.

To address this gap, this study focuses on how we can develop a semantic enrichment approach to support automated compliance checking applications in practice. The underground garage is one of the important facilities of an office building in central business districts and is much more expensive than parking above grade. Underground garages have strict requirements for space layout and fire safety, and developers tend to reduce the height of floors and the size of parking spaces to save on cost, which may lead to design errors and expensive rework. The current underground garage compliance checking mostly relies on manual review by experts, and the process lacks automation. The case of underground garage compliance checking is taken as an example, and in this case, we develop a computational geometric approach to enhance the semantics in BIM models. A prototype system is developed based on the proposed algorithms to illustrate the implementation of the semantic enrichment approach in real-world cases.

The remainder of this paper is organized as follows. The second section discusses related work. The third section illustrates the research method. The fourth section details the workflow of the approach and the algorithms used. The fifth section verifies the feasibility of the approach and analyzes the effectiveness of the developed prototype system in real-world cases, discusses the results obtained, and analyzes the limitations. The conclusions are given in the last section.

Related Work

BIM-Based Automated Code Compliance Checking

Code compliance checking aims to identify the defects and shortcomings of a specific design before construction transpires (Nawari 2020). In the current practice, the code compliance checking process is usually manually performed by domain experts such as architects and engineers, the manual process is time-consuming and prone to error. With the development of BIM technology, the data are presented in a machine-readable format, and some of the code checking processes have the potential to be automated to save time and provide reliable results. In the past few years, both scholars and industries are trying to establish automated code checking processes in different disciplines. Martins and Monteiro developed a prototype to perform the hydraulic analysis and the code checking in water distribution network projects (Martins and Monteiro 2013). Choi et al. developed a BIM-based evacuation regulation checking system for high-rise buildings (Choi et al. 2014). Demir Altıntaş and Ilal integrated BIM and GIS data to support automated code checking of building code at a neighborhood scale (Demir Altıntaş and Ilal 2021). In addition, some checking platforms (such as FORNAX, SMARTcodes, and Solibri Model Checker) have been developed to support some of the building code checking tasks (such as fire safety requirements environmental health requirements) of the government in Singapore and the United States (Eastman et al. 2009). Although there are already considerable automated code checking applications, it only covers a very small part of the building regulations. Most efforts on code checking applications are related to fire safety in buildings, but the possibilities of automation for other code checking tasks in practice have not been fully explored. Code checking for underground garages is an essential duty before construction. Design errors such as the noncompliance of

height clearance between pipelines and parking spaces can lead to expensive rework before a garage can be commissioned. But the current compliance checking process for underground garages is at a low level of automation, which entails a great deal of repetitive work and is prone to error.

BIM Semantic Enrichment

Various BIM applications have been developed to support AEC activities. The IFC, which is an open and platform-independent standard (ISO 16379), has been established for information interaction among different AEC applications. The IFC format provides the knowledge expression of various entities in the AEC field, provides a carrier for the information exchange among different platforms, and contains more than 600 entity definitions and 300 type definitions. For example, IfcBuildingElement is a generalization of all elements that participate in a building system, 20 subtypes are used to represent the main building elements (such as IfcWall, IfcDoor, and IfcBeam), and these building elements have their own attributes definitions. IfcRelationship defines the five kinds of relationship between objects (composition, assignment, connectivity, association, definition), and the relationships are usually used to provide semantics to support different applications such as indoor pathfinding (Fu et al. 2020). However, the IFC schema is not able to cover all building elements and relationships, while some applications like building performance analyses and code compliance checking require higher levels of semantics than the IFC schema can provide (Bloch and Sacks 2020). BIM semantic enrichment refers to the addition of valuable information that cannot be directly obtained in the original digital model (Belsky et al. 2016). At present, scholars have developed different BIM semantic enrichment approaches, and some machine learning-based approaches are developed to ensure the integrity of mapping BIM elements to IFC classes. Koo et al. extracted the attributes and relationships from the IFC file and used the support vector machine (SVM) algorithm as a classifier to detect misclassifications of eight kinds of architectural BIM elements (Koo et al. 2019). Similarly, Wu and Zhang developed a learning-based algorithm to classify building elements (such as beams, columns, footings, and slabs) in an IFC model into predefined categories (Wu and Zhang 2019). Geometric reasoning is often used to infer new semantic information from models, lower-level geometry primitives of the 3D model such as the face, polyline, and circle are often used to infer new semantics of objects. For example, pipes can be identified by curvature (Dimitrov and Golparvar-Fard 2015), doors and windows can be detected as the empty regions bounded by horizontal and vertical lines (Valero et al. 2016). Ma et al. applied experts' domain knowledge (including geometric features and pairwise relationships of objects) to classify bridge objects in 3D models (Ma et al. 2018). Although much effort has been carried on BIM semantic enrichment, the goal of most studies is to classify or identify objects to support the detection of misuse of BIM objects or 3D reconstruction tasks. Few studies have demonstrated the possibility of BIM semantic enrichment approaches in supporting engineering decision making in practice. In order to make a better understanding of how the semantic enrichment method can be used in engineering practice, this paper proposes a computational geometric approach that uses geometric information to enhance the semantics in BIM models to support underground garage compliance checking.

Research Method

To demonstrate how semantic enrichment methods can be used in practical applications, an example of underground garage

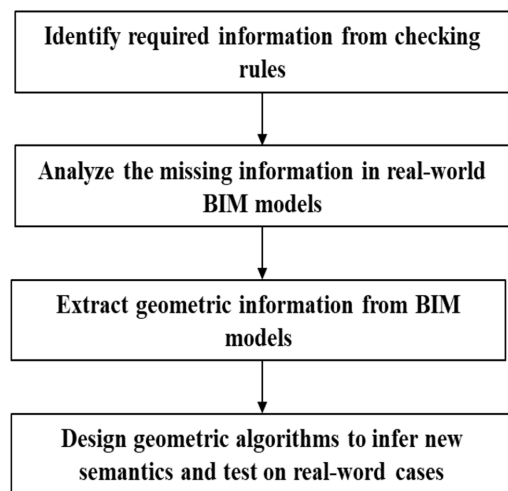


Fig. 1. Flowchart of the research.

compliance checking is selected. To meet the requirements for the underground garage compliance checking, this study develops geometric reasoning algorithms to enhance the semantics of the BIM models and uses real-world cases to demonstrate the effectiveness of the semantic enrichment approach for underground garage compliance checking. The framework of the research is shown in Fig. 1. The first step is to analyze the selected building codes and identify the required information for performing compliance checking. Then, compare the information required for compliance checking with the information that already exists in underground garage BIM models and figure out how designers model and find out what information is usually missing in most cases. Next, extract geometric information from BIM models and design geometric reasoning algorithms to enhance the semantics for supporting compliance checking. The final step is to test the approach on real-world cases and illustrate how the proposed approach can help automate the compliance checking process for underground garages.

BIM Semantic Enrichment Approach for Underground Garage Compliance Checking

The proposed approach aims to enhance the semantics in BIM models to support compliance checking for underground garages. After consulting a group of experienced designers in underground garage design, a set of meaningful clauses related to the code checking tasks for parking spaces, driveways, and ramps is selected to illustrate the general process of the proposed semantic enrichment

approach. The main clauses for compliance checking are as follows:

1. The slope of the ramp cannot exceed 15 degrees.
2. The minimum size of parking spaces. For example, the size of for perpendicular parking space shall be larger than 5.1 m × 2.4 m.
3. All parking garages shall have an unobstructed headroom clearance that is not less than 2.0 m for the ramp and 2.2 m for parking space and driveway to prevent collision with vehicles.

Based on these clauses, the tasks can be divided into three parts: (1) Identify ramp, parking space, and driveway, and other building elements in BIM models. (2) Get the attributes and obtain the shape extent of the ramp, parking space, and driveway. (3) Get the height difference among ramp, parking space, driveway with the above building component.

Then, real-world BIM models are analyzed to understand how designers model the underground garage in general cases and figure out what information has to be enhanced to support compliance checking for the selected clauses. The details of underground BIM models collected from five real-world projects are represented in Table 1. The modeling information related to the selected clauses is as follows. Driveways are modeled as a part of the slab in all models, the geometry of driveways cannot be directly obtained in the model for clear height calculation. Most projects use slab elements in the modeling software to create the car ramp of the underground garage, leading to semantic mismatch problems. Parking spaces are modeled as third-party components in the modeling software, but the geometric attributes of parking spaces are missing in the BIM models. To address these issues to support compliance checking, a computational geometric approach is proposed to enhance the semantics in underground garage models. The tasks of the semantic enrichment approach for underground garage models include elements identification (driveways and ramps), attributes inference (width, length, and slope), and geometric relationship inference (height clearance between different objects). The framework of the approach is shown in Fig. 2. The process begins with the extraction of geometric information from BIM models, then identifies elements in the underground garage and obtain the shape extent of elements, and after that, the implicit attributes and geometric relationships are inferred.

Extraction of Geometry Information from IFC

IFC files are exported from the BIM models of the underground garage. IFC model has multiple geometric representation methods and different local coordinate systems. In this study, we use an open-source tool ifcopenshell (<http://ifcopenshell.com>) to transform IFC geometries into OBJ format without understanding complex coordinate system and geometry definition in IFC schema.

Table 1. Overview of BIM models in the selected real-world cases

Project	Office building 1	Office building 2	Office building 3	Shopping mall 1	Shopping mall 2
Number of floors of underground garage	2	2	2	2	1
BIM authoring tool	Revit				
Driveway	Driveways are modeled as a part of the slab element, without a specific type definition. There are arrow elements (created by third-party components in the BIM authoring tool) on the driveways.				
Ramp	The designers use slab elements in the BIM authoring tool to model the ramp, the slope values have not been labeled.				
Parking space	The parking spaces are established by third-party components in the BIM authoring tool, the attributes such as length and width are missing.				
Other building elements	Building elements such as MEP components (e.g., sprinklers and ducts) and structural elements (e.g., wall, beams, and columns) are included.				

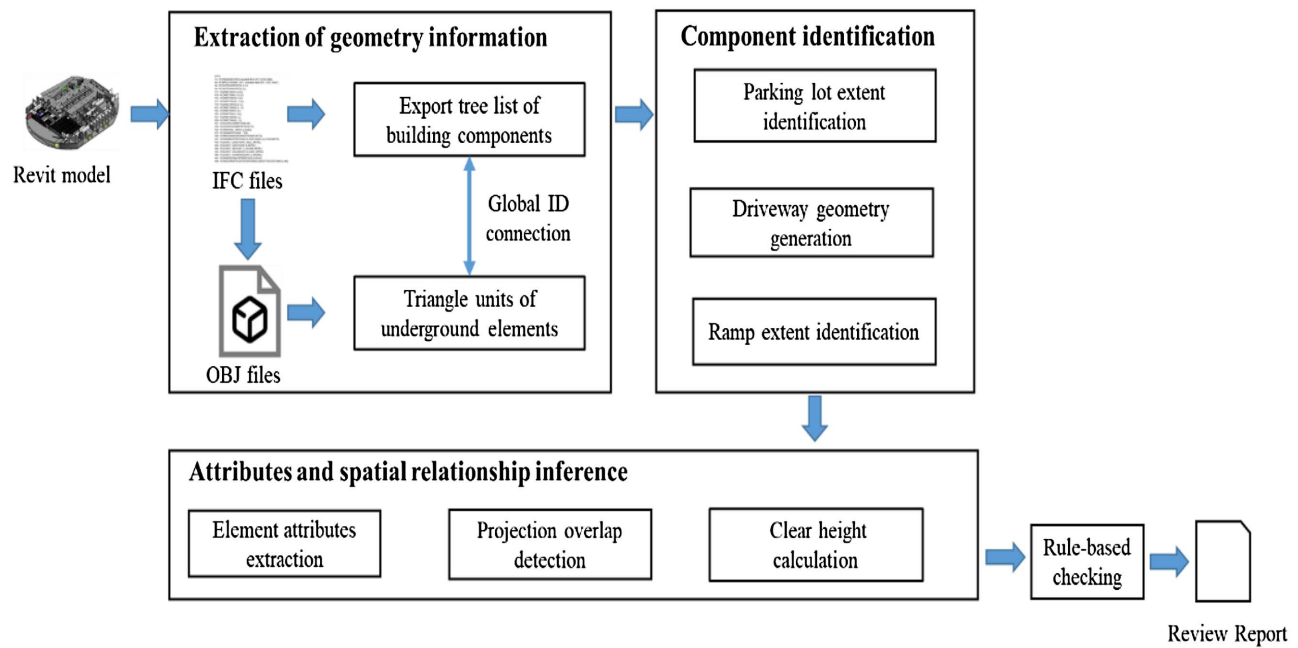


Fig. 2. Workflow of underground garage compliance checking.

Various types of 3D representations can be converted into a unified geometric representation, and the local coordinate system in the IFC format can be converted to a global coordinate system. In the OBJ format, the BIM geometric representation is converted into triangular meshes from Brep, CSG, and other representation methods in IFC. Each mesh has three vertices and is placed into a world coordinate system. In the compliance check task of the BIM model of the underground garage, some geometric attributes and topological information are not included in the IFC semantics. Geometric information in OBJ format is used to infer the missing semantic information to support underground garage compliance checking. Then, the Global ID of elements is used to link geometry in the OBJ model and the information in an IFC format. The geometric representation in OBJ format can bring benefits to geometric reasoning. For example, when calculating the clear height above elements in underground garages, some objects have different elevations at various locations, such as ramps and pipes, the elevation of the local position of the object can be obtained through the triangular mesh representation in the OBJ model.

Component Identification and Element Attribute Inference

After transferring the IFC file into OBJ format, the next step is to identify the shape extent of the objects. The Global ID for each building component can be used to link the geometry model in the OBJ with the product property in IFC. The range of X-coordinate and Y-coordinate of each object has to be identified at first. Then, the faces with overlapping projections must be selected on the XY plane, and the height difference between faces can be obtained by calculating the difference between Z-coordinates.

The equation of a face can be obtained by using the coordinates of three different points on the face. From the OBJ model in Fig. 3, we can obtain three vertexes (e.g., P_1 , P_2 , P_3) of a triangular face. The face equation is the basis for calculating the subsequent height difference and detecting overlap. The equation of a face is expressed as Eq. (1)

GUID	g 3J1u5z7Sr7AgKnKturzRWg
$P_1(x, y, z)$	v 22.4864974665445 73.705 -9.89694450390831
$P_2(x, y, z)$	v 22.4864974665445 74.205 -9.89694450390831
$P_3(x, y, z)$	v 21.9864974665445 73.955 -9.89694450390831

Fig. 3. Vertexes of triangular units.

$$AX + BY + CZ + D = 0 \quad (1)$$

where $A = (P_2 \cdot y - P_1 \cdot y) \times (P_3 \cdot z - P_1 \cdot z) - (P_2 \cdot z - P_1 \cdot z) \times (P_3 \cdot y - P_1 \cdot y)$; $B = (P_2 \cdot z - P_1 \cdot z) \times (P_3 \cdot x - P_1 \cdot x) - (P_2 \cdot x - P_1 \cdot x) \times (P_3 \cdot z - P_1 \cdot z)$; $C = (P_2 \cdot x - P_1 \cdot x) \times (P_3 \cdot y - P_1 \cdot y) - (P_2 \cdot y - P_1 \cdot y) \times (P_3 \cdot x - P_1 \cdot x)$; and $D = 0 - (A * P_1 \cdot x + B * P_1 \cdot y + C * P_1 \cdot z)$.

Parking Lot Extent Identification

In the IFC schema, a parking space is defined by `IfcBuildingElementProxy`, which served as a proxy definition of `Ifcproduct` to represent the undefined building elements in the IFC schema. It is customized by the designer and the element name can be used to identify the parking space, but the length and width of underground parking spaces are undefined in the IFC schema.

There are three types of parking spaces, namely, parallel, oblique, and perpendicular (Fig. 4). We find that the geometric elements that make up the parking spaces are often different in different BIM models; for example, the left parking space is modeled as a rectangular wireframe (Fig. 5), and the second is modeled as a rectangular face (Fig. 6).

To identify the range of different types of parking spaces and obtain the attributes, a general algorithm is designed as follows (Fig. 7):

1. Search all the vertices of a parking space in the OBJ file and select the max X-coordinate, min X-coordinate, max Y-coordinate, and min Y-coordinate from all of the coordinates.
2. Defined an axis-aligned bounding box (AABB) around the parking space by the extrema of the parking space mesh

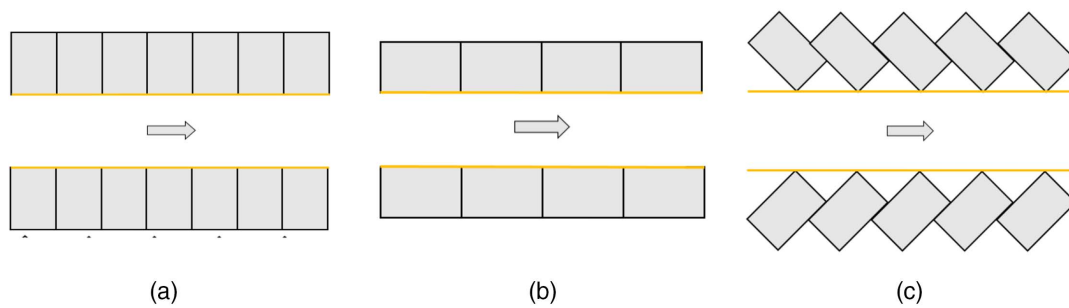


Fig. 4. Parking space with different orientations: (a) perpendicular; (b) parallel; and (c) oblique.

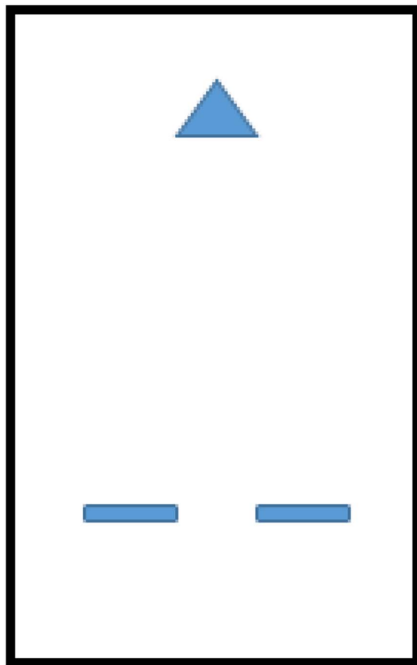


Fig. 5. Parking space represented by a rectangular wireframe.

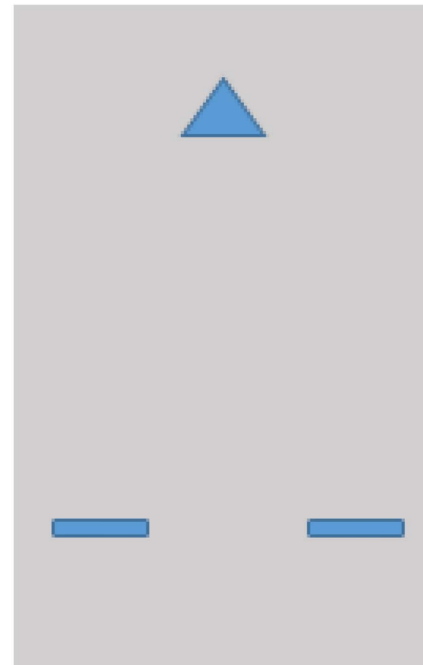


Fig. 6. Parking space represented by a rectangular face.

coordinates, namely, (X max, Y max), (X min, Y max), (X min, Y min), and (X min, Y max).

3. Select the points whose coordinates contain Y max, X max, Y min, and X min from a parking space geometry. These points refer to the vertexes of the parking space (e.g., P1, P4, P3, P2) in Fig. 8.
4. Divide a parking space into two triangular faces (e.g., Faces 1 and 2). Each face consists of three adjacent vertexes (e.g., P1, P2, P4 and P2, P3, P4), and the elevation of each face is denoted by the max z-coordinate among all points in a parking space.
5. Determine whether the parking space is parallel to the axis. If the X, Y-coordinate of a vertex is equal to the vertex of the bounding box, then the parking space is parallel to the axis; otherwise, it is the opposite. For example, in Fig. 8, if (P1. x, P1. y) is equal to (X min, Y max), then the parking space is parallel to the axis.
6. Property extraction. Calculate the Euclidean distance between two adjacent points, such as distance (P1, P2) and distance (P2, P3), and then select the longer one as the length of the parking space and the shorter one as the width.

After the aforementioned processes, the parking space is redefined with new properties, including name, Global ID, length, width, and the coordinates and elevation of the two faces.

Driveway Generation

In the BIM model, the driveway entities do not exist in the IFC schema, which is generally modeled as a part of the slab. In the underground garage design code, the driving direction must be marked on the lane to guide the vehicle. This study proposed a logic chain-based algorithm to infer the extent of the driveway from the position of the guide arrow. This algorithm begins from the arrow and spreads the rectangle. If the rectangle overlaps with an obstacle, then the generated rectangle is stopped from spreading and merged with the driveway. In this process, the AABB of the building elements on the floor (e.g., pillars, parking spaces, and walls) are considered to be obstacles.

The workflow of the driveway generation algorithm is presented in Fig. 9. This algorithm begins with the retrieval of arrows in the IFC file. Similar to the parking space, the arrow is also defined as IfcBuildingElementProxy, which can be recognized by its name. The Global ID in the IFC file is used to link to the OBJ model to find all the triangular meshes that make up the arrow. Then, the AABB of the arrows is obtained and the bounding box is extended by a certain distance (e.g. 1.5 m) along one direction of the short edge (negative y-axis) of the bounding box. Next, the rectangle is extended in both directions of the positive and negative y- and

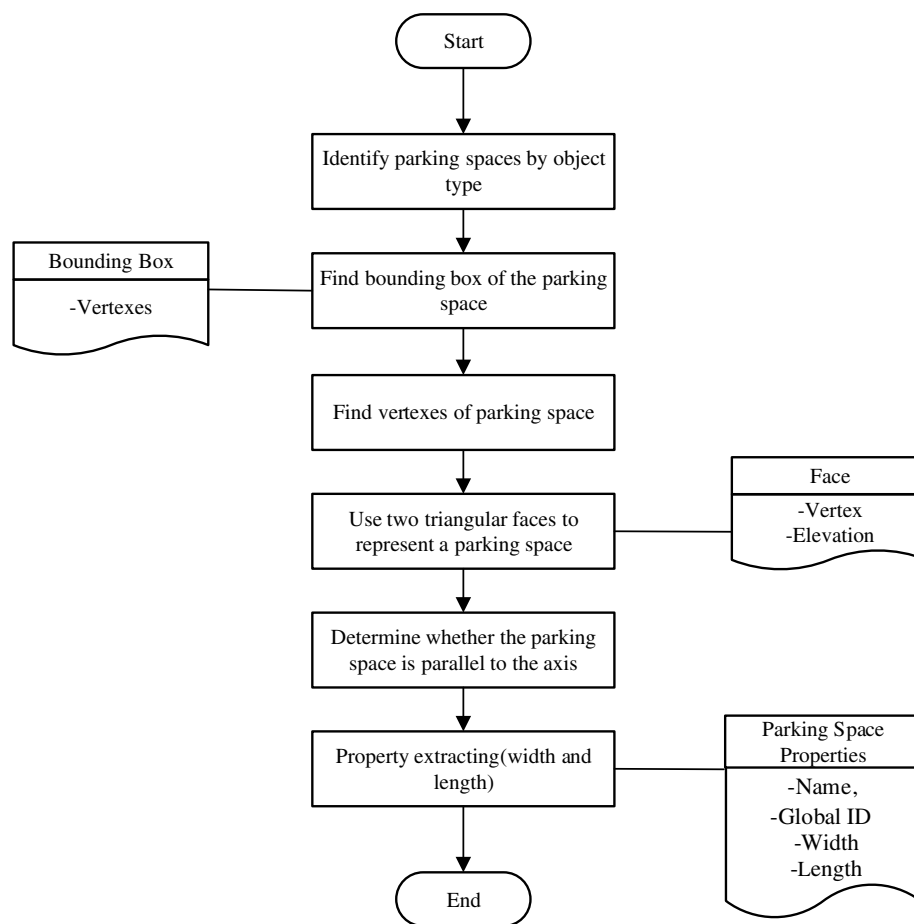


Fig. 7. Parking space identification and property extraction.

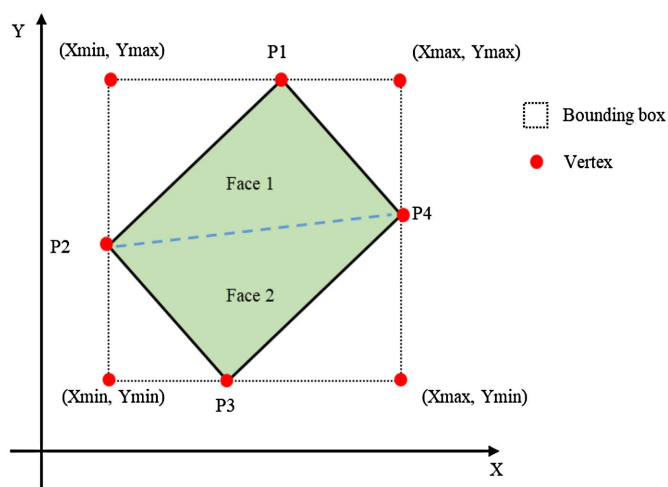


Fig. 8. Bounding box of a parking lot at any angle.

x-axes along the long side of the original AABB and along the short side of the original bounding box, respectively. This process is stopped when the bounding box of the arrow overlaps with the obstacles. Finally, all rectangles are merged to be the whole driveway in the underground garage.

The overlap detection algorithm used in the presented process is explained in the following section “Projection Overlap Detection.”

Fig. 10 shows the process of generating the straightway and the corner from arrows. Notably, the overlap detection is based on the bounding box of the object, the parking spaces, columns, and walls, which are projected onto the same plane. Subsequently, these objects are checked whether they intersect the projection of the lane. This algorithm is also applicable to oblique parking spaces because the AABB of the oblique lane in Fig. 10(a) is still parallel to the coordinate axis.

Ramp Extent Identification

In practice, we find that due to the complexity of modeling of spiral car ramp, many designers usually model it in the BIM model as IfcSlab instead of IfcRamp. The slope can be used to distinguish the ramp from the slab. However, it cannot be read in the property of IfcSlab. The geometric inference is performed to identify the slope of the ramp. A ramp geometry can be divided into hundreds of triangular meshes (Fig. 11). The maximum slope value of IfcSlab is selected by choosing each triangle unit and calculating its slope. When the slope exceeds a certain value, the area is identified to be a ramp.

Spatial Relationship Inference

In this research, the spatial relationship involved in this study refers to the clear height between components. The clear height calculation between two objects can be converted into the problem of determining the height difference among the overlapping parts of the two objects in the horizontal plane projection. Therefore, the clear

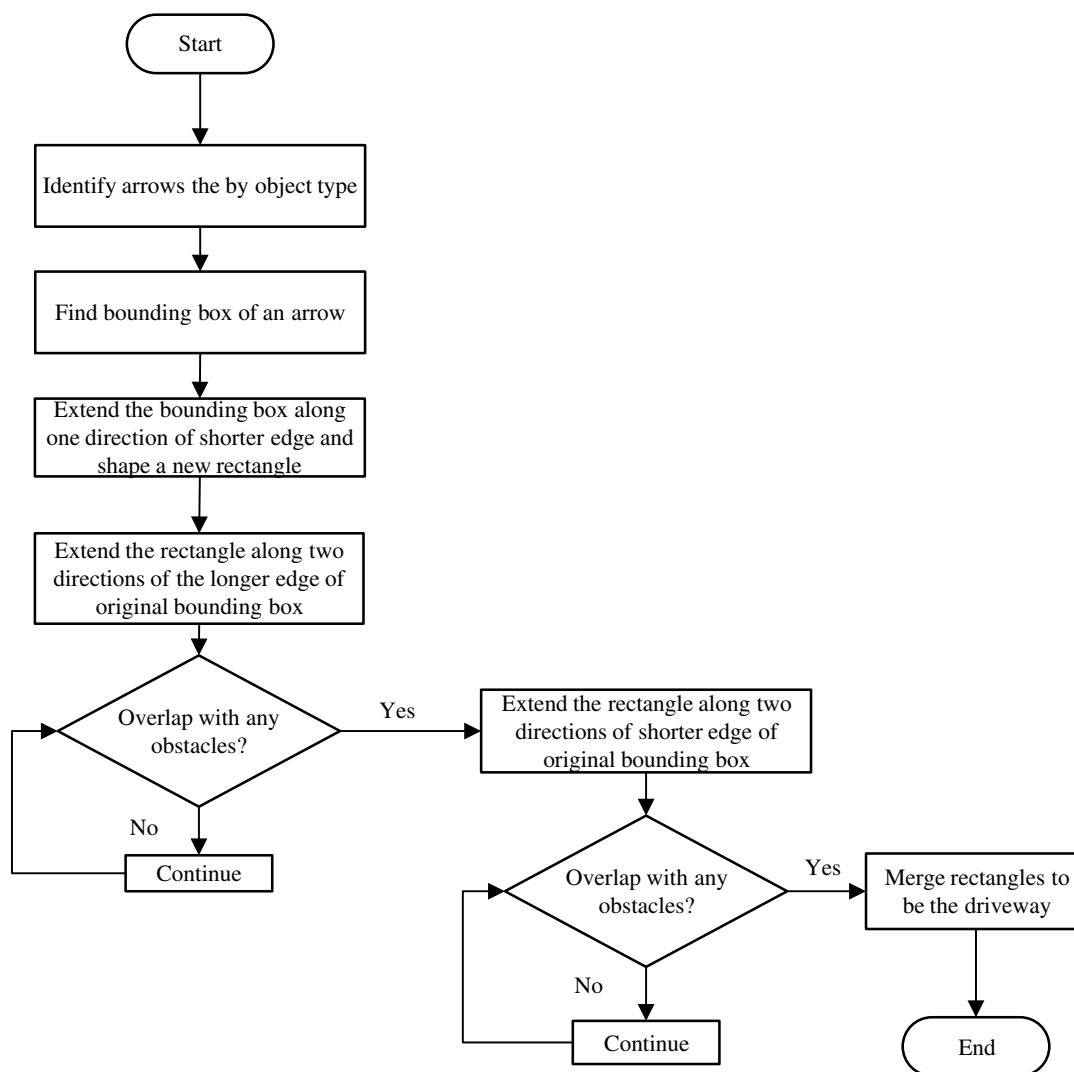


Fig. 9. Driveway generation algorithm.

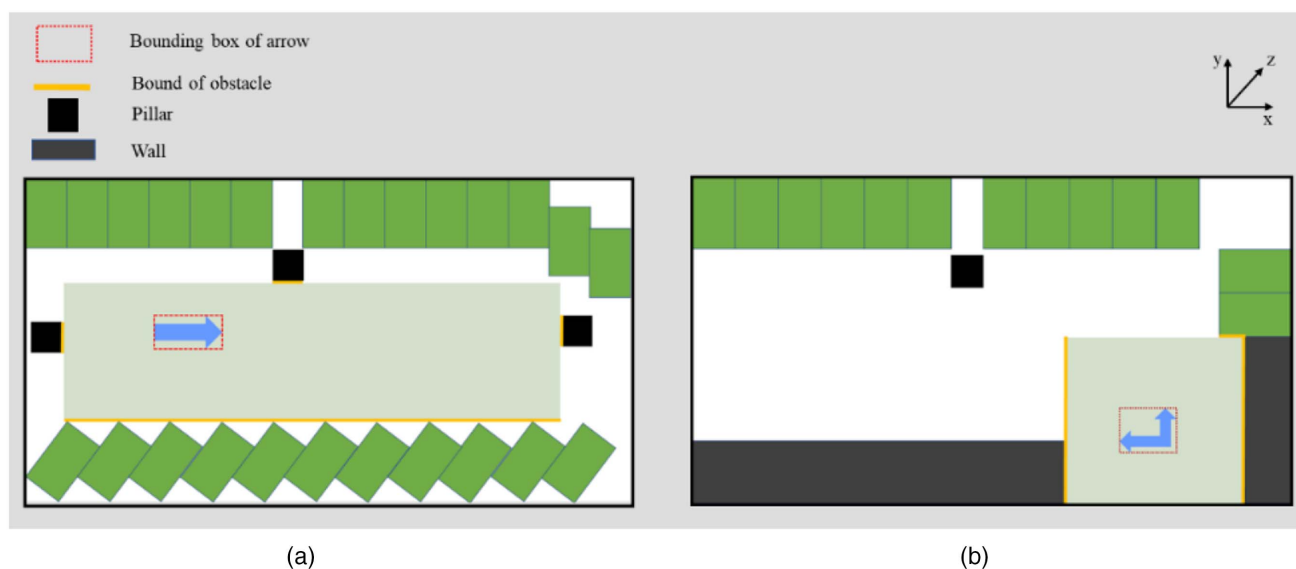


Fig. 10. Driveway generation: (a) straightway; and (b) corner.



Fig. 11. Meshes of the ramp.

height detection algorithm of the object is divided into two main steps. The first step detects whether the projections of the two objects on the XY plane overlap and excludes the objects whose bounding boxes cannot overlap. Subsequently, the triangular unit projection overlap detection is performed for the remaining objects, the triangle units of the two objects are projected onto the same plane to check whether there is overlap between the projections of triangle units that make up the two different objects. The second step refers to the height difference calculation of the projected overlapping parts. The flowchart of the overall process is shown in Fig. 12.

Projection Overlap Detection

Finding the projection overlap of the top and bottom components is preliminary work for calculating clear height. For underground components, the parking spaces and lanes include rectangles (Object C in Fig. 13), with four vertices that form the bounding box, and their geometry can be divided into two reorganized large triangular faces. The bounding box and the triangle face can be used to determine whether the object intersects with objects on the plane (for overlap detection in the driveway generation process) or the 3D objects' projection. The ramp is usually curved, but the bounding box of a triangle is also available from the max and min x and y coordinates of the triangle unit projection (Projection A in Fig. 13). For elements on the top, such as air supply duct, sprinkler, pipe, beam, and ceiling, and these elements are composed of tens to hundreds of triangular units in OBJ. The building element on the top produces a projection of the entire object (Projection B in Fig. 13) and a triangular unit (Projection A in Fig. 13) in the horizontal plane. The bounding box of the entire object projection is used to exclude objects that cannot be overlapped. This step is essential to speed up the entire checking process. The projection of the triangular unit that composes the object is used to determine which position of the object is not qualified.

Let A and B be the two geometries of the upper and lower objects, respectively. First, exclude disjoint rectangles by iterating through all vertices of each object, max X-coordinate, min X-coordinate, max Y-coordinate, and min Y-coordinate to form the bounding box of the object. The projections of A and B cannot overlap under four conditions (Fig. 14):

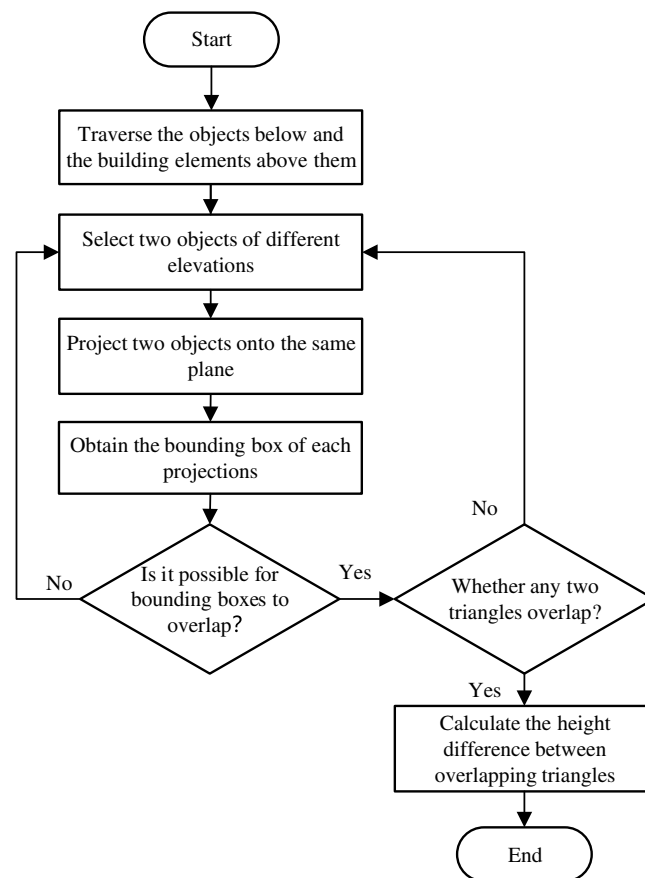


Fig. 12. Flowchart of clear height checking algorithm.

1. Projection of A is on the left: The max X-coordinate of A is less than the min X-coordinate of B.
2. Projection of A is on the right: The min X-coordinate of A is more than the min X-coordinate of B.
3. Projection of A is on the bottom: The max Y-coordinate of A is less than the min Y-coordinate of B.
4. Projection of A is on the top: The min Y-coordinate of A is more than the max Y-coordinate of B.

Then, the triangle units of the two objects are used to detect the overlap of projections. The method used for checking whether the triangle units of the projections of A and B are overlapping is as follows. Assume that six different points, namely, O, P, Q and R, S, T, while points O, P, Q are the vertices of a triangle OPQ and points R, S, T are the vertices of a triangle RST. If one of the vertices of triangle RST is inside triangle OPQ, then the two triangles are overlapping. Take point R as an example, the vector from point R to the vertices of triangle OPQ can be calculated as $\overrightarrow{OR}$, $\overrightarrow{QR}$, and $\overrightarrow{PR}$ (Fig. 15). Subsequently, T_1 , T_2 , and T_3 are the cross products of the aforementioned vectors using Eqs. (2)–(4)

$$T_1 = \overrightarrow{OR} \times \overrightarrow{QR} \quad (2)$$

$$T_2 = \overrightarrow{QR} \times \overrightarrow{PR} \quad (3)$$

$$T_3 = \overrightarrow{PR} \times \overrightarrow{OR} \quad (4)$$

If any condition in Eqs. (5) and (6) is satisfied in point R, then point R is inside triangle OPQ. Eqs. (5) and (6) are shown as

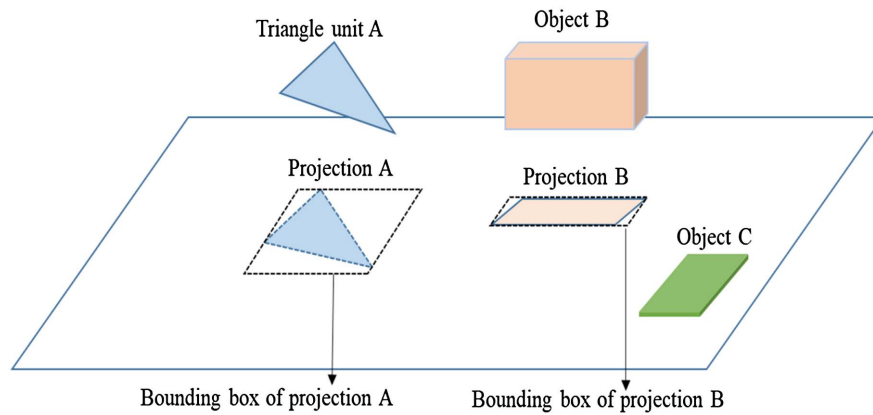


Fig. 13. Projection of objects and triangle units.

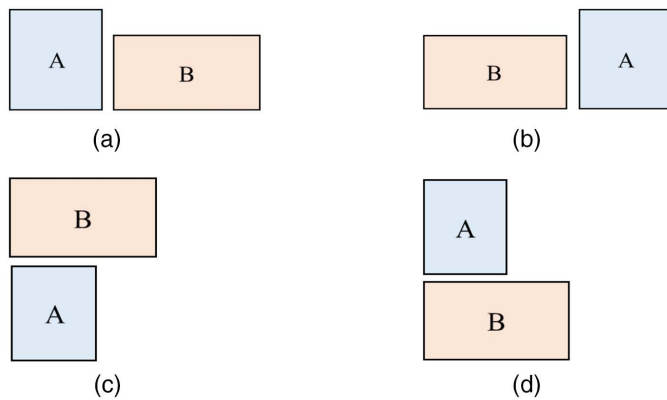


Fig. 14. Four situations wherein the projections of A and B cannot overlap: (a) projection of A is on the left; (b) projection of A is on the right; (c) projection of A is on the bottom; and (d) projection of A is on the top.

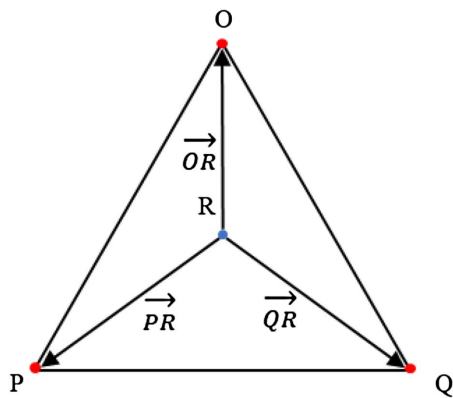


Fig. 15. Graphical representation of a point in the triangle.

$$T_1 > 0, T_2 > 0, T_3 > 0 \quad (5)$$

$$T_1 < 0, T_2 < 0, T_3 < 0 \quad (6)$$

The aforementioned process with points S and T is repeated. If any vertices of triangle RST satisfy Eqs. (5) or (6), then the triangle RST and OPQ are overlapping; otherwise, it is not overlapping.

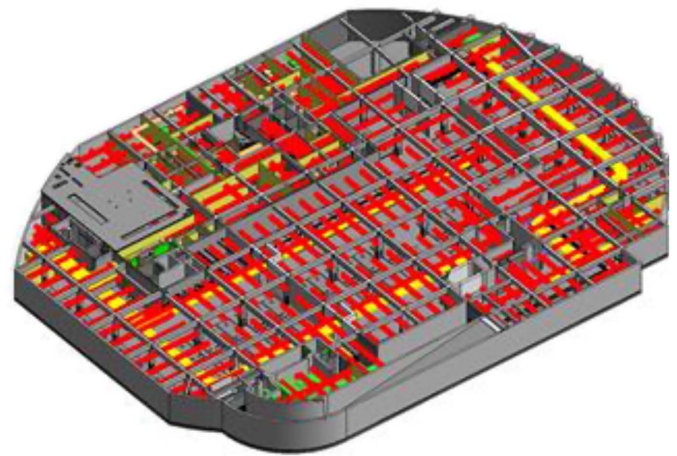


Fig. 16. BIM model of the underground garage (B2).

In this section, the bounding box is used to help eliminate cases where the projections do not overlap. For the remaining objects, a triangle overlap detection algorithm is applied to find objects with overlapping projections further.

Height Difference Calculation

Driveway and parking space are represented by two triangular faces. The elevation of the underground driveway and parking space is presented by the max z-coordinate among their triangular units. For the ramp, the elevation of each triangular unit varies, and a clear height calculation of all triangular units of the ramp is required. For the upper building elements, such as air supply ducts and sprinkler pipes, these components consist of tens to hundreds of triangular units, and the z-coordinate of different triangular units could vary. For each object that consists of triangular units with different elevations, the difference in elevations between the overlapping part of the upper objects and lane, parking space, and ramp store the minimum value as the clear height between two elements.

Case Study

Test cases are investigated to validate the proposed approach in this study. In this study, real-world BIM models are chosen and compliance checking is conducted based on the proposed semantic enrichment approach. An underground garage compliance checking

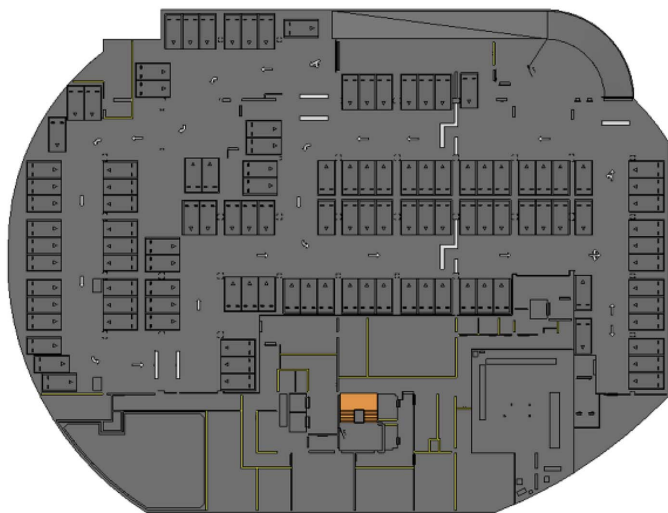


Fig. 17. Floor plan (B2).

prototype system is developed to show the potential of the semantic enrichment approach in practice.

Prototype Implementation

A prototype system is developed based on the proposed approach and tests in a real-world scenario demonstrate how the semantic enrichment approach benefits automated compliance checking for the underground garage. The prototype system has been developed as a web application, and the Unity platform is used to display BIM files on the web pages. The element information in the IFC model is parsed by the Java program. The geometries of the elements are extracted from the OBJ files and be stored in the Unity platform, and they are linked by global IDs of elements. Nine underground garage BIM models are selected to test the system, as we mentioned in the previous approach description section, Table 1 describes the characteristics of the input BIM models, all models have been manually exported as IFC instance files to be the input of the system. One of the models, which is a five-story

office building with a two-story underground garage, is selected as an example to illustrate the function of the system. The BIM model of the second floor underground is shown in Figs. 16 and 17. The air supply ducts, water pipes, spray pipes, and beams on the top of the garage increase the difficulty in controlling the clear height of the garage.

Initially, the Revit model is transferred into the IFC files while the IFC files of different disciplines (Architecture, Structure, MEP) are uploaded to the system. Subsequently, these files are automatically transferred into OBJ format. Fig. 19(a) shows the floor plan of B2. The proposed geometric reasoning approach is used for populating semantics that do not exist in the original BIM but are required during compliance checking. All of the parking spaces (Green) in this model are presented as rectangle faces, and the length, width, and range of each parking space are calculated. The driveway is part of the floor and has not been modeled separately, and no IFC class represents the driveway in the model. Then, the system will generate the 3D geometry of the driveway, load it into the original 3D garage model (Fig. 18), and display the driveway (Blue) on the Unity web page. Then, the height difference is calculated to check if any locations do not meet the clear height requirement. The code checking of the length and width of parking spaces are performed automatically, and the ramp slope checking is also included in the proposed system. The unqualified components will be highlighted in the system (turn red) [Fig. 19(b)]. Finally, the system will output the review report (Fig. 20).

Testing of Prototype

The prototype was tested on real-world examples, the semantic enrichment task in this research faces varying degrees of complexity. First, nine real-world models were used to demonstrate the effectiveness of the lane generation algorithm. Lane generation is the most difficult task because lane elements do not exist in the BIM model. Among the nine models, two real underground garages of large public buildings were taken as examples. This study verified the effectiveness of the algorithm by comparing the lane marked by the expert and the lane generated by the algorithm. These two models have the largest area and complex layout. The first underground garage is 103 m long and 140 m wide, with an area of approximately 14,000 square meters. This underground garage is part of

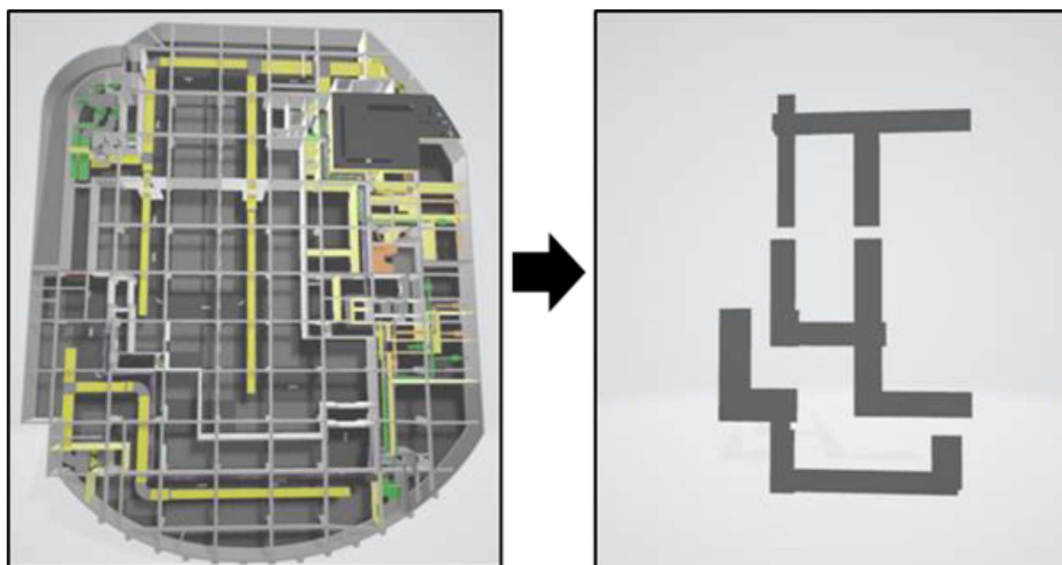


Fig. 18. Generated OBJ model of the driveway.

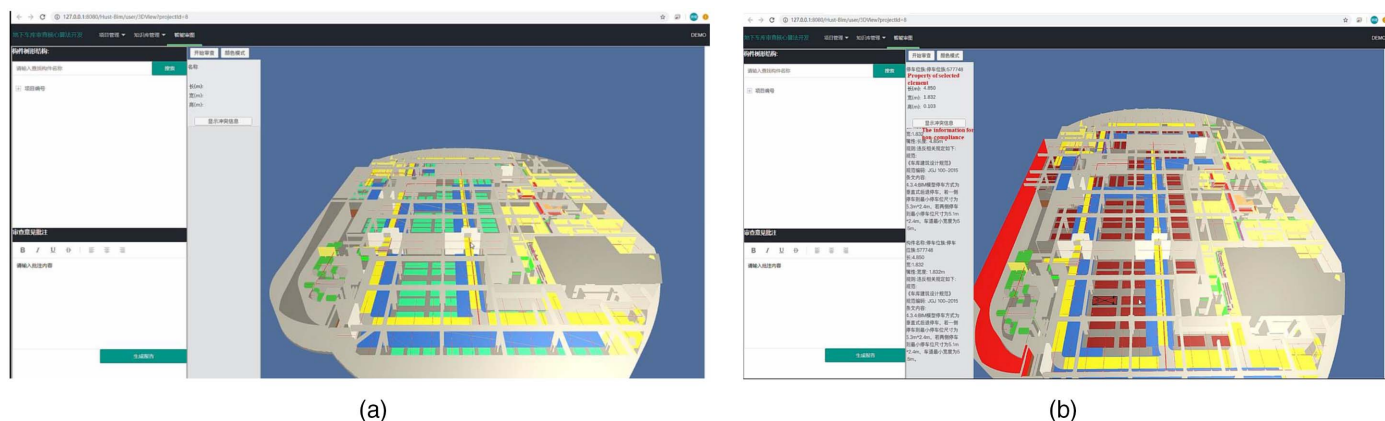


Fig. 19. (a) Driveway generation in the system; and (b) unqualified components (red color) displayed by the system.

Design Review Report BIM 模型合规性审查意见告知书		
项目基本信息 Project Information		
项目名称: XXXX 地下车库	送审编号: 20191316	
送审时间: 2020-03-19	送审单位: XXX 大厦	
建设单位: XXX 大厦	勘察单位: XX 勘察设计院	
设计单位: XX 建筑设计院	审查完成时间: 2020-03-29	
未通过审查规范条文列表及审查意见 Review Results		
受建设单位委托, 本审查机构对报送的 BIM 模型进行合规性审查。		
审查范围: 地下车库		
审查参考规范: 1: 《车库建筑设计规范》(JGJ 100-2015) Referenced Building Code		
审查软件输出不合规结果如下: 共有 142 处违反相关规定 A total of 142 non-compliant components		
1、 审查构件名称: 停车位族: 停车位族: 577590 Element: Parking-space-ID577590 冲突构件名称: 族 2: 580557 Conflicting Element: An element customized by the designer-ID577590 建模不规范-构件重叠 Reason: Intersection		
2、 审查构件名称: 停车位族: 停车位族: 579105 Element: Parking-space-ID579105 冲突构件名称: 矩形弯头 - 半径 - 法兰: 1.0 W: 751841 Conflicting Element: Flange (Air supply duct)-ID751841 违规属性: 车位净高 Unqualified Attribute: Clear height of parking space 属性值: 1.607m Attribute Value: 1.607m 违反相关规定如下: 规范: 《车库建筑设计规范》(JGJ 100-2015) Referenced Building Code 条文内容: 4.3.6: 停车位及车道最小净高均为 2.20m. Clause Violated: The clear height of the parking space needs to exceed 2.2m		

Fig. 20. Report generated by the system.

a large shopping mall [Fig. 21(a)]. The second underground garage is part of a large office building with an area of more than 7000 square meters [Fig. 21(c)]. Figs. 21(a and c) represent the lanes on the BIM model labeled by human experts. Figs. 21(b and d) represent the lanes generated by the proposed algorithm. The red boxes represent the parts of the lanes that were not generated completely. In Fig. 21(b), all lanes were generated completely. In Fig. 21(d), only a very small part of the lane was missing, and most of the lanes were completely generated. Except for these two cases with more complex lanes, the success rate of lane identification on the other models is also similar to manpower. We use the ratio of the total length of the lane (generated by the algorithm) to the total length of the lane (marked by experts) to represent the recognition success rate. In the nine real-world models, the average recognition success rate in the tested models was 98.3%. By analyzing the results of the tests, we find that a low recognition success rate usually occurs when the lane width changes. Fig. 22 shows the area with missing lanes in the second model. The lane in this area has many

turns, and the lane width changes five times. Walls and pillars are observed on the lane as obstacles, the arrows are used to generate the lanes, and the red boxes represent the missing part of the lane. The obstacles are not in the same straight line. Thus, the lane width changes many times, thereby resulting in incomplete lane generation. In the proposed algorithm, the arrow first spreads along the long side of the bounding box and then spreads along the short side of the bounding box. If the arrow spreads along the short side first and then spreads along the long side, then the generated lane will be different. We can take the union of the lane geometries generated by these two methods to obtain the complete lane, thereby making the algorithm more accurate but reducing its speed.

We also compared the other checking items between the system with a group of experts (three), and the results for the other semantic enrichment tasks are presented as follows. 1) Attributes inference (width, length, and slope) and element identification (ramp): compared with experts judgment results, the slope of each ramp and the length and width of three kinds of parking spaces (perpendicular, parallel, and oblique) were all obtained correctly from the system, and all of the ramps were correctly recognized from floor elements in BIM models based on the slope value. For the presented tasks, the accuracy for attribute inference and element identification reaches 100%, which is at the same level as human experts. Fig. 23(a) shows the case scenario where the parking space width is less than the value required by the clause, and the value is inferred by the proposed geometric reasoning algorithm. 2) Geometric relationship inference (height clearance between different objects). Based on the successful identification of driveways and ramps, the height clearance above parking spaces, driveways, and ramps has been calculated. All elements that were detected to violate the clear height requirement and overlap were manually scrutinized in the BIM model, and the unqualified components can be divided into two types. The first type corresponds to the components with a clear height greater than 0 but less than the standard value (e.g., 2.2 m for a driveway). These elements violate clear height requirements. Fig. 23(b) shows the scenario where the clear height between the duct and the lane is less than the standard value. The second type corresponds to element overlap, which is classified as a modeling error. The clear height is 0, and the cause of this error is due to some components that are modeled to intersect with lanes, parking spaces, or ramps. Figs. 23(c and d) show the intersection between the parking space and structural components. Noncompliance of the clear height requirement detected by the system was compared with the judgment of the experts, and the system detected that the clear height of 16 locations exceeds the standard value (exclude modeling error) in the nine BIM models, while

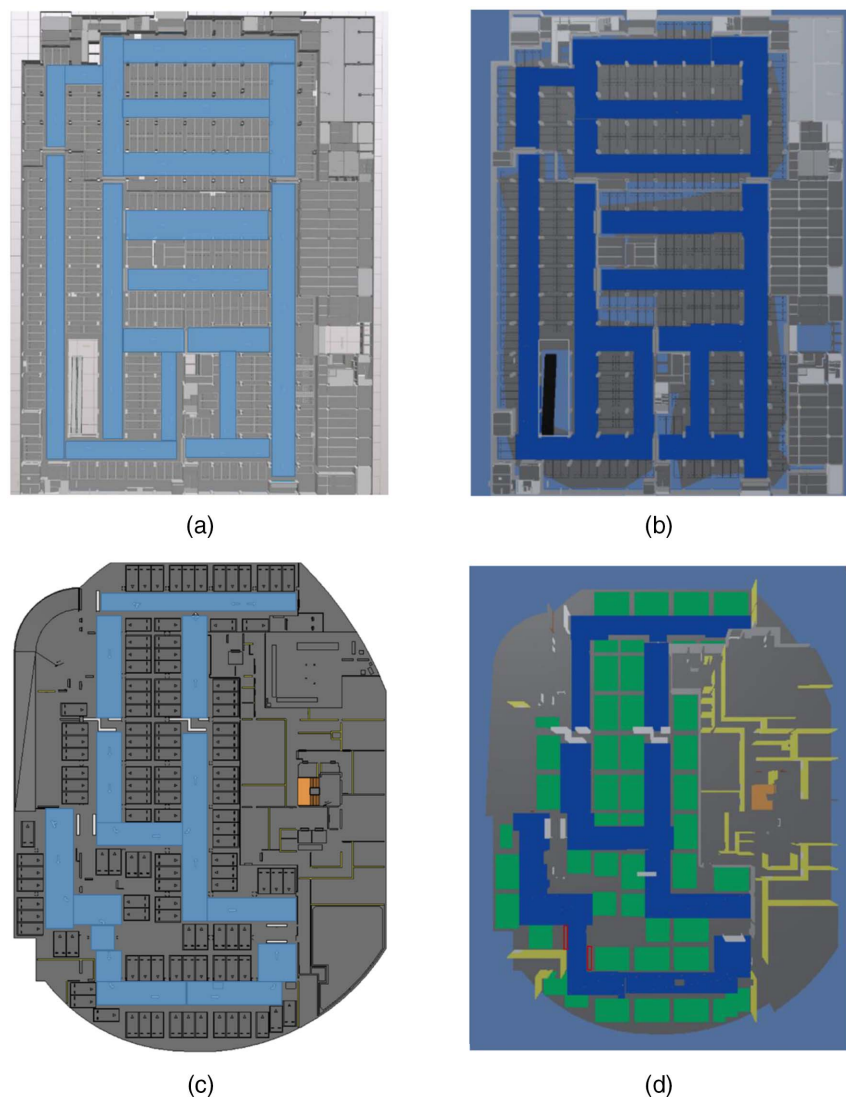


Fig. 21. (a) Lanes labeled by human experts; (b) lanes generated by the algorithm; (c) lanes labeled by human experts; and (d) lanes generated by the algorithm.



Fig. 22. One of the complex areas in the underground garage model.

experts only found 12 locations; thus, the system performs better than experts on the clear height checking task.

Discussion

The case study presented in this work demonstrates that the proposed computational geometric approach can identify missing semantic information to support compliance checking for underground garages. Tests in real-world cases have proven that the developed system based on the proposed approach has the potential to improve the degree of automation in real-world compliance checking tasks. Different from other BIM review work, this system can populate missing semantics that do not exist in the original BIM automatically. To meet the operating requirements of the system, the users only need to name the parking spaces and arrows correctly and put them in the same location as the actual building, modelers do not need to add a lot of additional semantic information to support compliance checking, which is not in line with modeling habits.

The prototype system developed in this work uses triangular mesh representation to infer missing information in the BIM model, each geometry is divided into hundreds of triangular units.

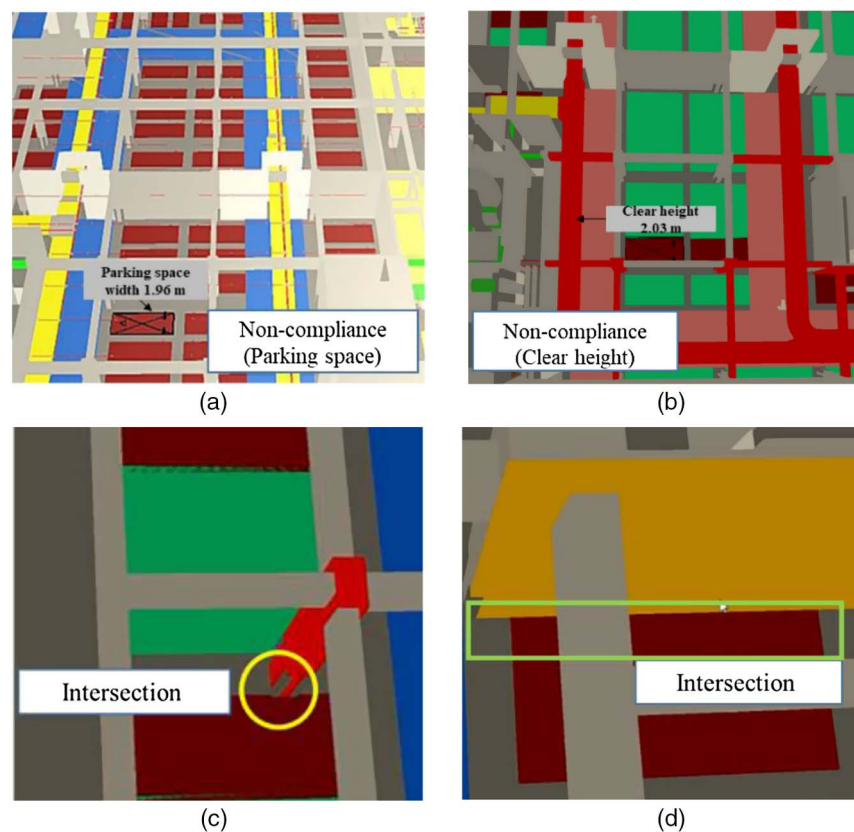


Fig. 23. Different unqualified situations: (a) parking space width (1.96 m) is less than the standard value (2.1 m); (b) clear height (2.03 m) is less than the standard value (2.2 m between the driveway and the air duct); (c) intersection between a column and parking space; and (d) intersection between a wall and parking space.

Each vertex has its coordinates, and the review can cover every position on the building component and avoid omissions through the proposed approach. The clear height detection method proposed in this work is also suitable for detecting the clear height of other spaces, such as rooms and stairs. The geometric reasoning approach can identify missing lanes in the BIM model, as well as identify parking spaces and ramps and infer the attributes, based on the geometric information provided by OBJ format.

The proposed approach also has some limitations. For example, in some situations, such as when the lane width changes frequently, our algorithm cannot fully recognize the lane. Moreover, the driveway generation algorithm is mainly applicable to the recognition of horizontal and perpendicular lanes because the edges of the AABB of the arrow are along the coordinate axis direction, and the generated lane is also along the coordinate axis direction. All the nine models tested in this work are horizontal or perpendicular. Although this situation is common in the underground garage design, diagonal lanes still exist in some real-world cases, and the diagonal lane may cause incomplete recognition. Future research will further explore the identification of arrow directions and path-finding algorithms to solve the problem of oblique lane identification. In this study, only nine BIM models are tested. Thus, more tests on real-world models will help improve our algorithm to cope with arbitrary input from designers.

Conclusions

This study aimed to provide a demonstration of how can semantic enrichment approach apply to support engineering decision making

in real-world cases. Thus, this study developed a computational geometric approach to enhance semantics in the BIM model to support automated compliance checking for underground garages. This study first identified the required information referenced in building code related to the underground garage and then summarized how designers model in most situations from real-world cases and figured out what information should be enhanced in the BIM model to support underground compliance checking. Afterward, geometric reasoning algorithms were designed based on the observation of characteristics of the BIM models. Finally, a prototype system was developed to test the effectiveness of the proposed approach in real-world cases. The results show that the proposed method in this study can effectively enhance semantics in underground garage BIM models for supporting code compliance checking tasks.

The main contribution of this study is two-fold. First, a computational geometric approach is developed to enhance BIM semantics in this study, and the prototype implementation and test results have shown the potential of the semantic enrichment approach in real-world cases. Second, the proposed approach helps improve the level of automation in underground garage compliance checking practice, and it can be adopted by practitioners to solve problems in real-world cases.

Furthermore, the proposed algorithms can detect other 3D elements with overlapping projection and can be applied to other clear height review tasks in the BIM model. In this study, the clauses are currently read by humans and structured in the knowledge base. In future research, natural language processing can be studied further to manage the rule base to address more complex compliance checking tasks.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

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